Weak response of warmer growing seasons on tree growth across 11 species of adult trees in an urban garden

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Introduction

Anthropogenic climate change affects many different biological processes with one of the most documented being shifts in phenology—the study of recurring life history events (Cleland *et al.*, 2007; Menzel *et al.*, 2006; Parmesan *et al.*, 1999). Earlier springs and later autumns have lengthened the window of opportunity during which plants can grow (Duputié *et al.*, 2015). For trees, longer growing seasons means they should grow more, which is especially significant given the critical role that forests play in offsetting human-emitted greenhouse gases (Keenan *et al.*, 2014; Stridbeck *et al.*, 2022; Harris *et al.*, 2021). For example, in temperate forests, the 1% largest trees store half of the living biomass, which is largely composed of carbon sequestrated from the atmosphere (Lutz *et al.*, 2018; Mensah & Petersson, 2024). Given that temperate forests account for 40% of the global carbon sink (Gibbs *et al.*, 2025), this makes adult trees disproportionately important in the carbon dynamics of these ecosystems. The future of that carbon sink, however, largely relies on whether these trees will keep growing with longer growing seasons.

The long standing assumption that longer growing seasons should increased growth has recently been challenged by studies that failed to find this relationship (Dow *et al.*, 2022; Silvestro *et al.*, 2023; Čufar *et al.*, 2015; Charlet De Sauvage *et al.*, 2022; Cabon *et al.*, 2022). Recent research suggests that warmer springs—that largely contribute to extending the growing season—have not increased the maximum tree growth rate or annual increment in temperate mixed forest communities (Dow *et al.*, 2022). Similarly, Čufar *et al.* (2015) showed no growth increment in relation with the growing season length for *Fagus Sylvatica*. However, neither of these studies could clearly establish why they observed this decoupling. This highlights the necessity to understand why trees may not grow more with longer growing seasons.

Tree growth strategies and wood anatomy may shape tree species responses to growing seasons. Life-history strategies certainly affects trees' ability to track changing conditions. Longer growing seasons may benefit slower growing species because appropriate conditions would allow them to keep their low growth rate active for a longer period (Cuny *et al.*, 2012). In contrast, faster growing species can complete their season growth increment more quickly, making extra growing days less important (Cuny *et al.*, 2012). Now, wood anatomy in angiosperm trees impacts their growth temporality with ring-porous species starting to grow several days or weeks before new leaves emerge and before diffuse-porous species growth (Hessl *et al.*, 2026; Savage *et al.*, 2022; D'Orangeville *et al.*, 2022). Thus species with different wood anatomy experience considerably different temperature regime during the growing season; this results in an asymmetric effect of the season length between these species. Wood anatomy is inherently a strategy by itself. The reason ring-porous species start growing before new leaves emerge is because of their large xylem vessels that go through embolism during winter, and thus, the water transportation pathway needs to be restored. These large vessels make these species more vulnerable to drought and late spring frosts, but allow them to source water deeper into the ground (Taneda & Sperry, 2008; Kitin & Funada, 2016). In contrast, diffuse-porous species have smaller vessels that are less likely to embolize and thus tend to leafout earlier in the season than ring-porous species. However, there is diverging evidence of a direct correlation between wood porousness and life-history strategies with Lv *et al.* (2025) proposing that diffuse-porous species are slower-growing and longer-living than ring-porous species, while Taneda & Sperry (2008) proposed the opposite. While the growth rate of a species is directly related to its wood anatomy or not, species with these traits still exhibit distinct strategies on their own with allocation to storage or growth, being one of theme.

How trees store the carbon they accumulate (whether it is through a direct investment in growth or for long-term storage) could mute short term responses to longer growing seasons (Hessl *et al.*, 2026). The previous year thermal conditions can affect the following year's growth. Deciduous trees often use stored carbon at the beginning of the growing season because they need to grow new leaves before starting photosynthesis (Monson *et al.*, 2018; Wolkovich *et al.*, 2025). Mature trees in particular can rely on carbon that was stored for several years for their early season growth (Hessl *et al.*, 2026). Undeniably, environmental conditions during the current year directly influence growth. However, these carbon allocation strategies may blur annual growth responses because of a multi-year lag effect of carbon allocation to storage.

Here, we will look at the growth and growing season length relationship of mature trees in an urban arboretum. For this, we used 11 species of mature trees across nine years and looked at the growing season length and thermal conditions during that period. We also investigated the effect of the previous year thermal conditions on the current year growth. We used growth with tree rings and leaf phenology for the start and end of season that citizen scientists collected.

Methods

Citizen science program

The Tree Spotters is a citizen science program started in 2015 that aims to train citizen scientists for accurate and rigorous phenological monitoring at the Arnold Arboretum of Harvard University (42.30 °N, -71.12 °W; Figure 1). From 2015 to 2024, hundreds of citizen scientists monitored 77 individual trees and shrubs of 14 species. They regularly followed individuals from budburst in the spring to leaf coloring in the fall using the National Phenology Network (NPN) phenophases (Denny *et al.*, 2014): leaves (483), colored leaves (498), fruits (516), ripe fruits (390), falling leaves(471), recent fruit or seed drop (504), increasing leaf size (467), breaking leaf buds (371), flowers or flower buds (500), open flowers (501), pollen release (502). Not all phenophases were recorded for every tree, for every year, and some trees are missing several years of data.

Here we use leafout to colored leaves (or leaf coloring) to define the growing season length. Taken together, our dataset comprises 321 leafout and 284 colored leaves observations (which we refer to as the 'full datasets'), but we did not have both observations for all trees in all years, so our total of matched full growing seasons duration (which we refer to as the 'restricted dataset') is 278 observations. We visually checked for SOS, EOS and GSL outliers and verified that there was no observation bias potentially caused by a particular observer or in a particular year.

Phenological monitoring and sample collection

From 20 to 22 April 2025, we collected tree ring samples only for the tree species that were previously monitored for phenology—excluding *Fagus grandifolia* because of beach bark disease concerns—for a total of 49 individuals (Table S1). Thus, we sampled four or five individuals for 11 species: Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*). For this, we collected two 5-mm diameter cores, 15-cm length at 1.3 meters above ground, perpendicular to the slope and at 180 degrees from each other, using an increment borer (Mora Borer, Haglöfs Sweden, Bromma, Stockholm, Sweden). We cleaned the increment borer with alcohol (70% ethanol) and the inside with a brush before collecting each core. We stored the cores in modified plastic straws to promote aeration and left them to dry at ambient temperature for three months.

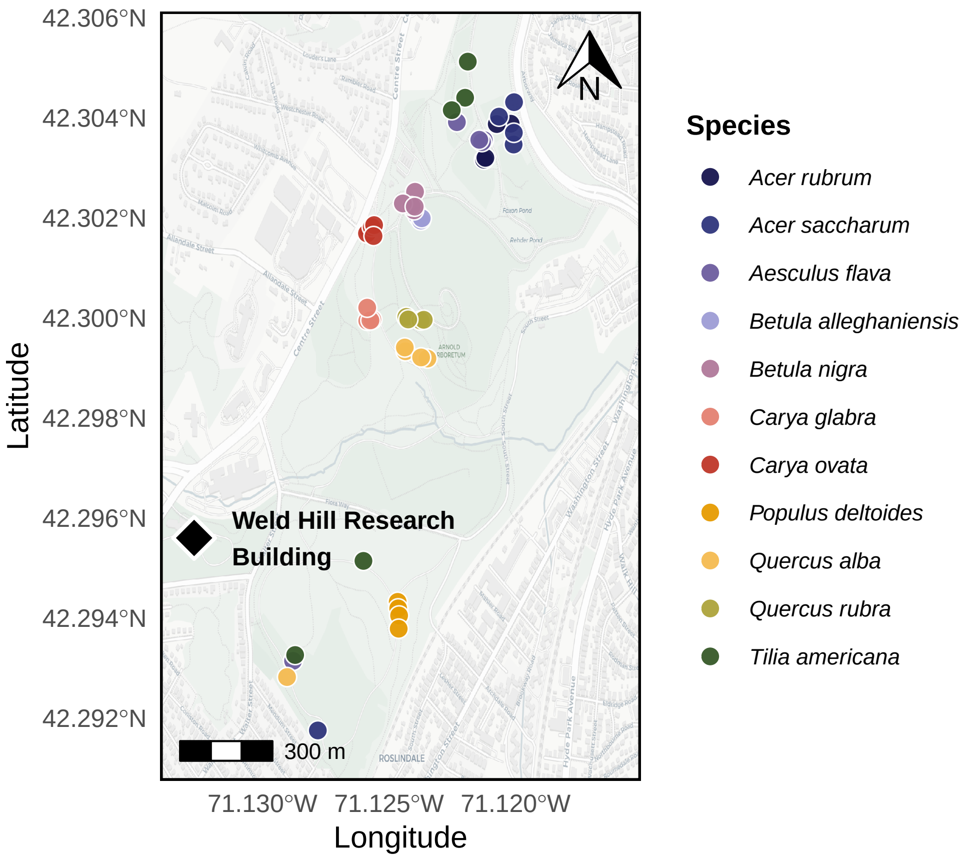

Figure 1: Map of the Arnold Arboretum of Harvard University with the locations of the trees depicted as the dots. The colors of the dots match the model estimates of model estimate figures (e.g. Figure 2). We also show the location of the Weld Hill Research Building (black diamond), where the common garden from Chapter 1 took place.

Sample processing, imaging and measuring

We used wooden mounts to stabilize the cores that were then sanded using progressively finer sandpaper grit (from 150 to 1000) by increments of roughly 150. We then used a high-resolution scanner to scan the cores at a resolution of 6250 dpi (Fong *et al.*, 2026), and used those images to measure the ring widths using ImageJ (Schneider *et al.*, 2012).

We visually cross-dated the ring width chronologies by assigning a calendar year to each ring and comparing annual variation of trends across the different individuals of the same species (Cook & Kairiukstis, 1990). Our short chronologies and low species replication prevented us from performing statistical cross-dating. We also did not detrend our chronologies because the ring widths during our study period (nine years) showed no consistent allometric trend (Figure S1). We included year in our models to control for effects of different years.

Data analyses

We used Bayesian hierarchical models to understand how growth shifted with four different season metrics: GDD, GSL, SOS and EOS. We used growing degree days (GDD; mean daily temperature above 5°C) between leafout and leaf coloring to calculate the thermal season and the number of days between leafout and leaf coloring for the calendar season (growing season length; GSL). Then, we used leafout for the start of season (SOS) and leaf coloring for the end of season (EOS). To better illustrate ring width responses to our season metrics, we report ring width change (log(mm)) in response to the number of GDDs accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD) for the thermal season. Similarly, we report ring width change per week of season length, leafout and leaf coloring for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero across all models.

We combined daily climate data from two sources located next to the Arboretum for the duration of our study (2016-2024). From 2016-2019, we used the Weld Hill research station climate station (Arnold Arboretum, 2026) and the "Boston (Weld Hill)" station for the remaining years (2020-2024; Carroll *et al.*, 2026). We determined drought occurrence at the arboretum using the North American Drought Monitor Maps and used the proposed drought index to qualify their intensity (National Centers for Environmental Information (NCEI), 2026).

Thus for each season predictor, we estimated its effect on ring width (i, denoting each observation) observed for each tree (*treeid*) in each year (*year*) as:

$$\begin{aligned} ln(ringwidth)_i &\sim Normal(\mu, \sigma_y) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}}) \end{aligned} \tag{1}$$

With this model, we estimated a species-specific effect of each season predictor on growth ($\beta_{species}$), but we also controlled for variation due to species ($\alpha_{species}$), individual tree (α_{treeid}) and year (α_{year}). We first ran the models using natural units and then scaled the predictors through z-scoring so we could directly compare their effect sizes (Gelman & Hill, 2006). We report estimates as means and 90% uncertainty intervals (UI). We also fit the models for SOS and EOS using the full vs restricted datasets. Finally, because the wood growth onset is synchronized with a different leaf phenophase for species with different wood anatomy (ring-porous being more closely synchronized with budburst and diffuse-porous, with leafout), we ran the models using budburst instead of leafout as the SOS. This allowed us to test whether our model responses diverged across diffuse vs ring porous species (Table S1).

We used an additional model to account for the previous year thermal conditions. For this, we used the GDD accumulated between leafout and leaf coloring from the previous year (e.g. for growth year of 2020, we used the GDD of 2019 as the previous year GDD), alongside the GDD accumulated during the current year. For interpretability, we also report ring width change (log(mm)) as the number of GDDs (for both years) accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD). Thus, we estimated the previous and current year thermal season effect on the current year's growth as:

$$\begin{aligned} \ln(\text{ringwidth})_i &\sim Normal(\mu, \sigma) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} \\ &\quad + \beta^{\text{current}}_{\text{species}} \times \text{GDD}_{\text{current}} + \beta^{\text{previous}}_{\text{species}} \times \text{GDD}_{\text{previous}} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \tag{2}$$

While controlling for the same variation across species, individual tree and year, this model allowed us to account for the effect of the thermal season during the current year ($\beta^{\text{current}}_{\text{species}}$), but also during the previous year ($\beta^{\text{previous}}_{\text{species}}$).

Finally, we used a third Bayesian hierarchical model to determine whether the SOS and EOS of our species were related during the current year. For this, we estimated the effect of the SOS on the EOS on each tree, in each year as:

$$\begin{aligned} \mathrm{EOS}_i &\sim Normal(\mu_i, \sigma_y) \\ \mu_i &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \mathrm{SOS} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \tag{3}$$

Results

Our dataset comprised 11 deciduous tree species with leaf phenology data spanning nine years (2016 to 2024) for a total of 321 leafout and 284 leaf coloring observations, yielding 278 complete growing seasons (GS) for 49 individuals. Throughout the duration of the study, mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C. 2020 and 2022 had moderate droughts in the summer (June, July and August), which ramped up to severe in September. Additionally, a moderate drought in June 2016, ramped up to severe in July, then extreme in August and September. The calendar season (GSL) ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021), while the thermal season (GDD) ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016). The earliest start of season (leafout) was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020), while the earliest leaf coloring day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016). Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019; Figure S3).

Across our eleven species and season predictors, we did not find any clear response of the growing season length on growth. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth 'RW/day' or 'RW/GDD' with days and GDD being adjusted to represent more interpretable units (see 'Data analyses' section in Methods for more details on predictor adjustment). Longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.08, UI: -0.12, -0.03 RW/GDD; Figure 2a and e; Table S2). Further, a longer calendar season did not change growth for any species (Figure 2b and f; Table S2). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar seasons were similarly predictive (see Figure S11.

Previous year climate also had limited effects on growth. For the one species that showed decreased growth with warmer thermal season, this effect persisted for the current year, but previous year GDD actually increased growth for *B. alleghaniensis* (β = 0.15, UI: 0.02, 0.28 RW/GDD; Figure 3b; Table S5). *B. alleghaniensis*'s congeneric, *B. nigra*, in contrast grew less with previous year GDD (β = -0.08, UI: -0.17, -0.00 RW/GDD; Figure 3b; Table S5) and current year GDD increased growth for that species (β = 0.12, UI: 0.03, 0.21 RW/GDD; Figure 3a; Table S5). Interestingly, *B. nigra* did not solely respond to current year's conditions (Figure 2a and e; Table S2), but grew more with warmer thermal season only when the previous conditions were considered.

An earlier start of season did not increase growth for any species and decreased growth for *T. americana* (β = 0.15, UI: 0.02, 0.27 RW/day; Figure 2c and g; Table S2), which did not respond to the thermal or the calendar season. Additionally, a later end of season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.09, UI: -0.16, -0.01 RW/day; Figure 2d and h; Table S2), which also decreased growth with warmer thermal season. Then, the estimates were unchanged when using the full or restricted datasets (SOS: n = 321 and EOS: n = 284 vs n = 278; Figure S9b), or when using budburst instead of leafout as the start of the growing season (Figure S10c).

Finally, a later SOS delayed EOS only for *P. deltoides* (β: 1.04, UI: 0.39, 1.70 EOS/SOS; Figure S12). We did not find any clear difference in baseline growth across years (Figure S2; Table S3) and between species (Figure S6; S7). Growth was similar across individual trees (Figure S8).

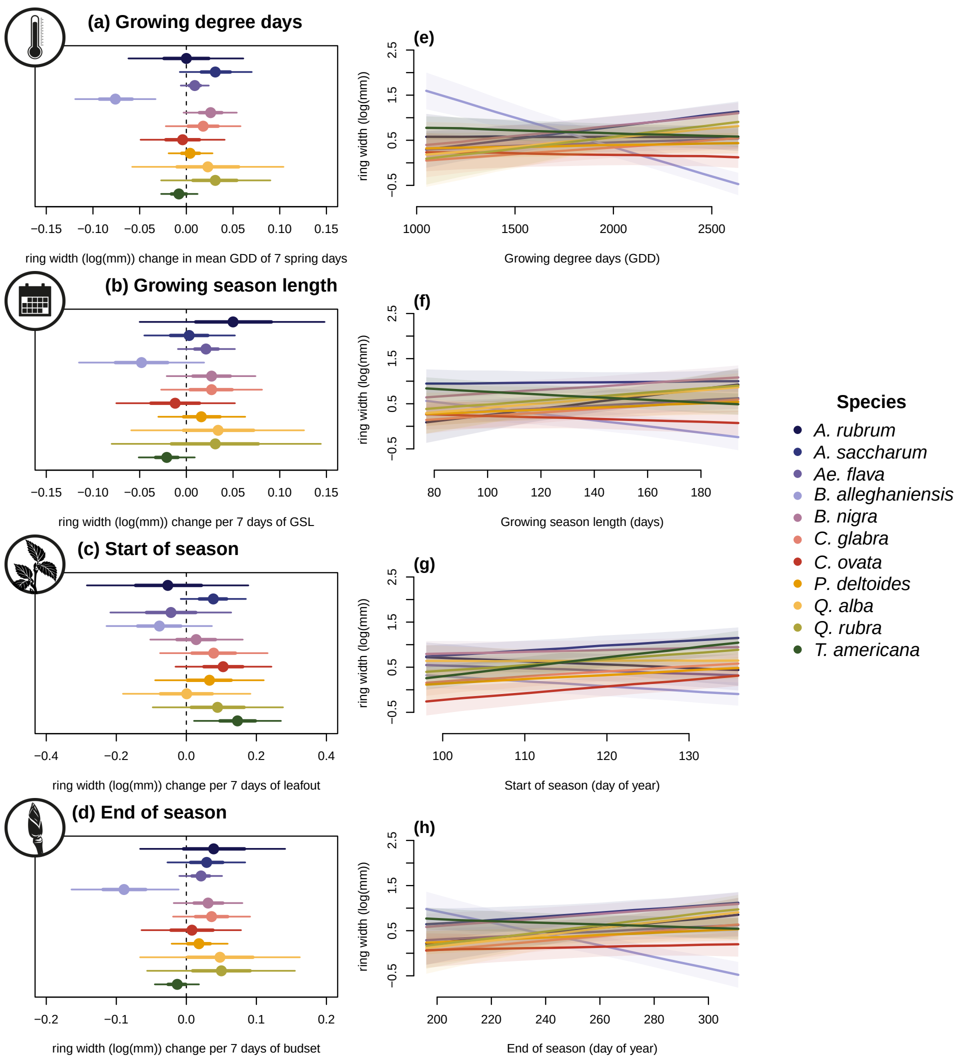

Figure 2: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length; start of season; end of season across 11 species: (Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*)); species colors are the same across the panels). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). See Table S2 for parameter estimates. Panels on the right (e to h) show the ring width response curves as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals. These models do not include an effect of the previous year GDD (see Figure 3 for these estimates).

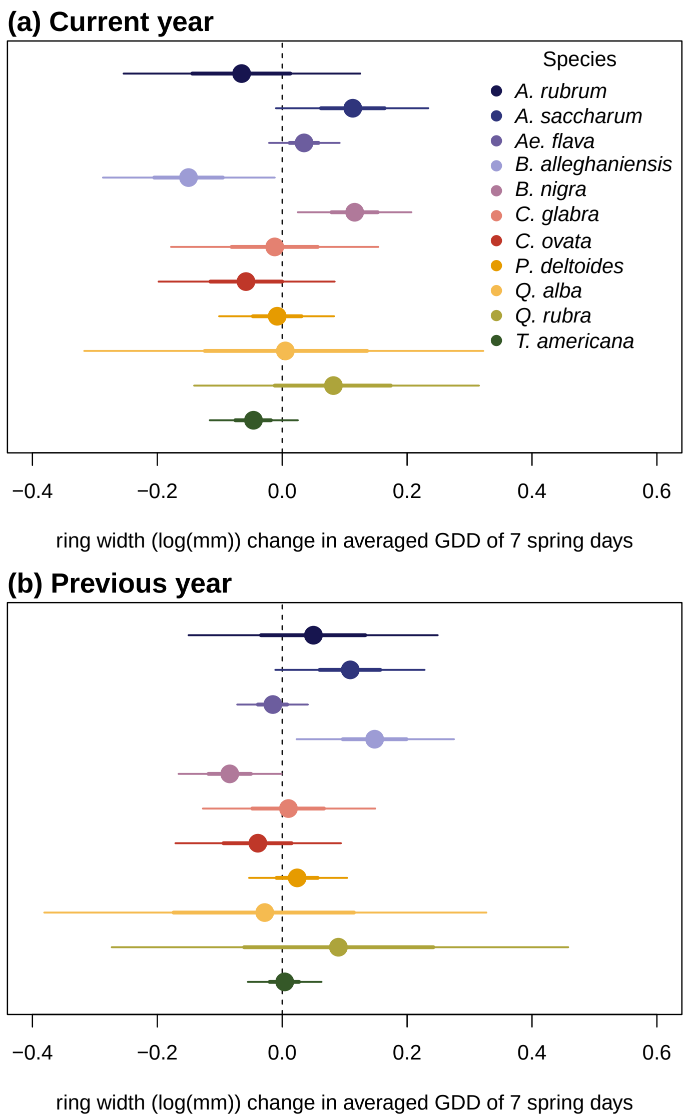

Figure 3: Model slope estimates representing ring width change (log(mm)) as a function of the current year growing degree days (GDD; a) and the previous year GDD (b). GDD represent the accumulated growing degree days (mean daily temperature above 5°C)) between leafout and leaf colouring and the models estimates shows ring width change (log(mm)) per scaled GDD (see 'Data analyses' section in Methods). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

Discussion

In this study, we used 49 trees of 11 species to understand if mature trees grew more with longer growing seasons (GS). To do this, we used annual growth with tree rings and leaf phenology for the SOS and EOS. Our findings reveal that warmer and longer GS did not increase growth for any species and decreased growth for one. We also found that the previous year thermal conditions shifted growth trends for two species.

Our findings support recent results that trees do not grow more with longer growing seasons. The most comparable study (with eight species in common to our study) was conducted geographically close to ours and found similar results. Despite warmer and earlier springs, trees did not grow more. Our results nicely complement these findings as we used leaf phenology and have EOS and SOS, while they only had SOS for growth phenology and not leaf phenology (Dow *et al.*, 2022). In addition, Čufar *et al.* (2015) had substantially more years (five decades) and found analogous results to ours, but only for one European species, *Fagus Sylvatica.* Then Cabon *et al.* (2020) showed a decoupling between gross-primary production and tree ring increments at the global scale, but obviously did not account for species differences. Taken together, these results support the idea that longer growing seasons may not increase temperate tree growth, a least for mature trees.

We previously showed (see Chapter 1) that our juvenile trees grew more with longer and warmer growing seasons, contrasting the response from our mature trees. Interestingly, these diverging responses corroborate the assumption that younger trees are usually more responsive to changing conditions than mature trees (Wolkovich *et al.*, 2025). While these species were all relatively fast-growing, our mature trees did not show strong life-history strategies in their growth response to longer seasons, with even the fastest growing (i.e. *B. nigra*, *P. deltoides* and *T. americana*) barely showing a response to warmer seasons.

Life-history influence on growth trends

While our species share an array of growth strategies (see Table S1), we did not find a consistent growth trend across them. Only *B. alleghaniensis*, which has a moderately fast-growth, responded to the thermal season and its decrease in growth suggests it may be negatively impacted by warmer seasons. *B. alleghaniensis* also grew less with a later EOS, which is a period characterized by fast GDD accumulation. The fact that *B. alleghaniensis* appears to be negatively impacted by warmer seasons suggest that this species could face local extirpation at its southernmost range because growth would not keep up with warmer and longer seasons (Doak & Morris, 2010; Aitken *et al.*, 2008). In opposition, *B. alleghaniensis*'s faster growing congeneric species, *B. nigra*, increase in growth during the current year—only when the previous year conditions were included—confirms the expectation from diffuse-porous species of a short-term response in rapid growth. Additionally, the fast-growing *T. americana* (Table S1; Burns *et al.*, 1990) grew less with an earlier SOS—when the days are short and cool. This is contrast with our expectation of species with this life-history strategy which should provide them an edge by starting to grow earlier in the season (Loughnan *et al.*, 2025). The other eight species had different life-history strategies (Fast: *P. deltoides*; moderate: *A. rubrum*, *A. flava*, *C. ovata*, *Q. rubra*; slow: *A. saccharum*, *C. glabra*, *Q. alba*), and despite this diversity in growth speed, neither changed their growth with different season drivers. Ultimately, these findings suggest a weak signal of life-history strategies in growth responses to the GS length and thermal conditions. Though, we found that these trends shifted slightly when considering also the previous year thermal conditions.

Growth response blurred by carbon allocation to storage

Shift in results when including previous year effects suggests including carbon allocation strategy in how we understand life-history strategies may be critical. The growth response of two *Betula* species shifted when we included the previous year thermal season as a driver. These two congeneric species (*B. alleghaniensis* and *B. nigra*) had diametrically opposed response trends, however. *B. alleghaniensis* growth decreased with the current year thermal conditions, but increased with a warmer previous year, possibly revealing a carbon allocation strategy that prioritizes carbon storage above short-term growth. This is in contrast with the expectation dfrom iffuse-porous species that tend to allocate less carbon to storage and more to growth (Barbaroux & Breda, 2002; Muhr *et al.*, 2016). In opposition, *B. nigra* increase in growth during the current year—only when the previous year conditions were included—confirms the expectation from diffuse-porous species of a short-term investment in growth with less allocation to storage. These opposite responses of two closely related species, highlight the importance of considering more than the current year in tree growth and GS length models.

Also, considering not solely the current year conditions when studying mature tree growth could help elucidate tree growth dynamics as even fast-growth species may have their responses blurred by carbon allocation to something else than growth. Our findings reveal a weak response of the GS length and conditions on growth across species, corroborating previous findings on mature trees (Čufar *et al.*, 2008; Delpierre *et al.*, 2016; Silvestro *et al.*, 2023). Most of our studied species did not grow more with a warmer thermal or longer calendar season. However, out results emphasize the importance of considering the previous year conditions in future growth and GS length studies because of the vast array of carbon allocation strategies that may blur short-term tree growth responses. GS are expected to warm and lengthen with, but it is still dubious how mature trees will track this trend. Whether tree growth stalls or decrease, both responses will influence the contribution of temperate forest to act as a global net carbon sink.

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Supplemental

Supplemental Methods

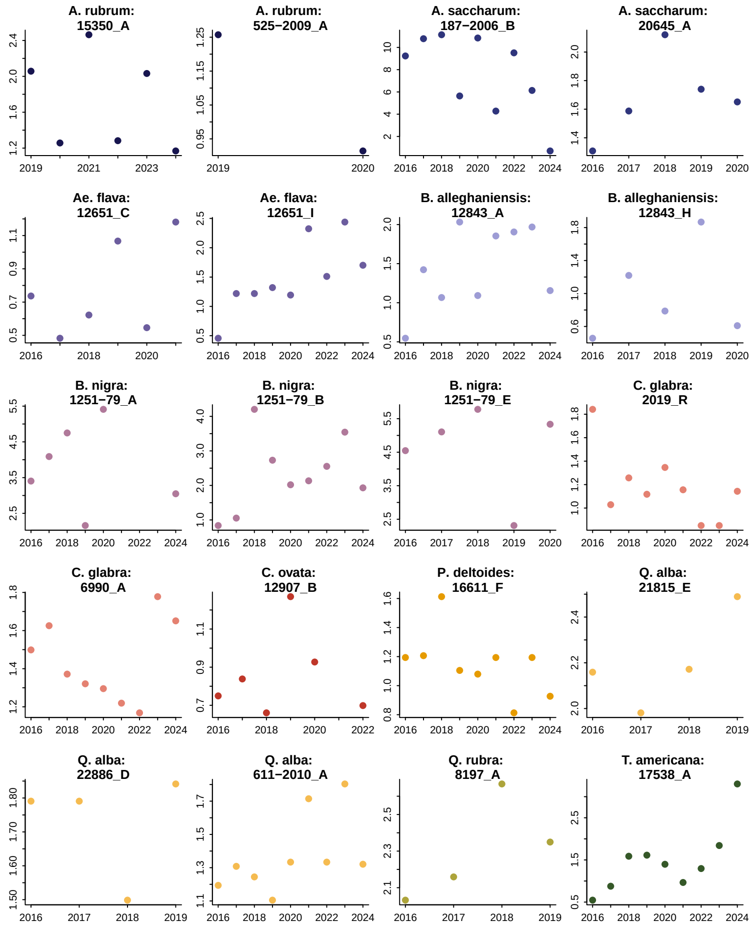

Figure S1: We randomly sampled 20 trees from our dataset and represent the ring width as a function of year to represent the allometry patterns across all years we had phenology data for.

Table S1: Treespotters species grouped by tree type, growth speed, and wood anatomy. We collected the growth speed information for each species from Burns *et al.* (1990)We sampled a total of 49 individual trees

Common Name (Latin binomial)	Growth speed	Wood Anatomy	n
American basswood (*Tilia americana*)	Fast	Diffuse-porous	5
Eastern cottonwood (*Populus deltoides*)	Fast	Diffuse-porous	4
Northern red oak (*Quercus rubra*)	Moderate	Ring-porous	4
White oak (*Quercus alba*)	Slow	Ring-porous	5
Pignut hickory (*Carya glabra*)	Slow	Ring-porous	4
Shagbark hickory (*Carya ovata*)	Moderate	Ring-porous	5
River birch (*Betula nigra*)	Fast	Diffuse-porous	4
Yellow birch (*Betula alleghaniensis*)	Moderate	Diffuse-porous	4
Sugar maple (*Acer saccharum*)	Slow	Diffuse-porous	5
Red maple (*Acer rubrum*)	Moderate	Diffuse-porous	4
Yellow buckeye (*Aesculus flava*)	Moderate	Diffuse-porous	5

Supplemental results

Table S2: Species level slope parameters which represent ring width change (log(mm)), per scaled predictor (see 'Data analyses' section in Methods for more details on predictor scaling). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). Table S3 reports all the models parameters.

Species	GDD	GSL	SOS	EOS
A. rubrum	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
A. saccharum	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
Ae. flava	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
B. alleghaniensis	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
B. nigra	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
C. glabra	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
C. ovata	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
P. deltoides	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
Q. alba	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
Q. rubra	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
T. americana	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

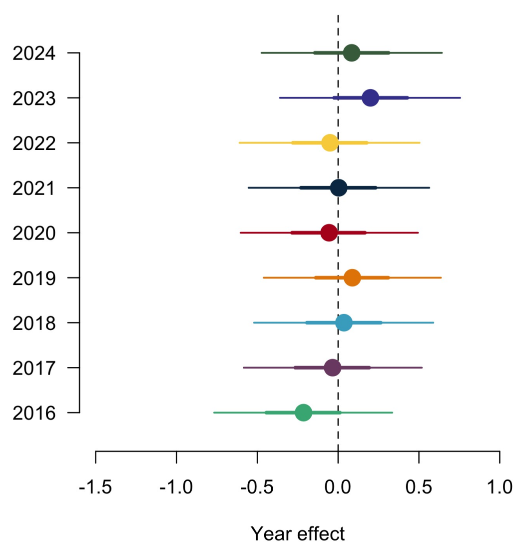

Figure S2: Ring width (log(mm)) deviations from the grand mean for each year using the GDD model (Table S3). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

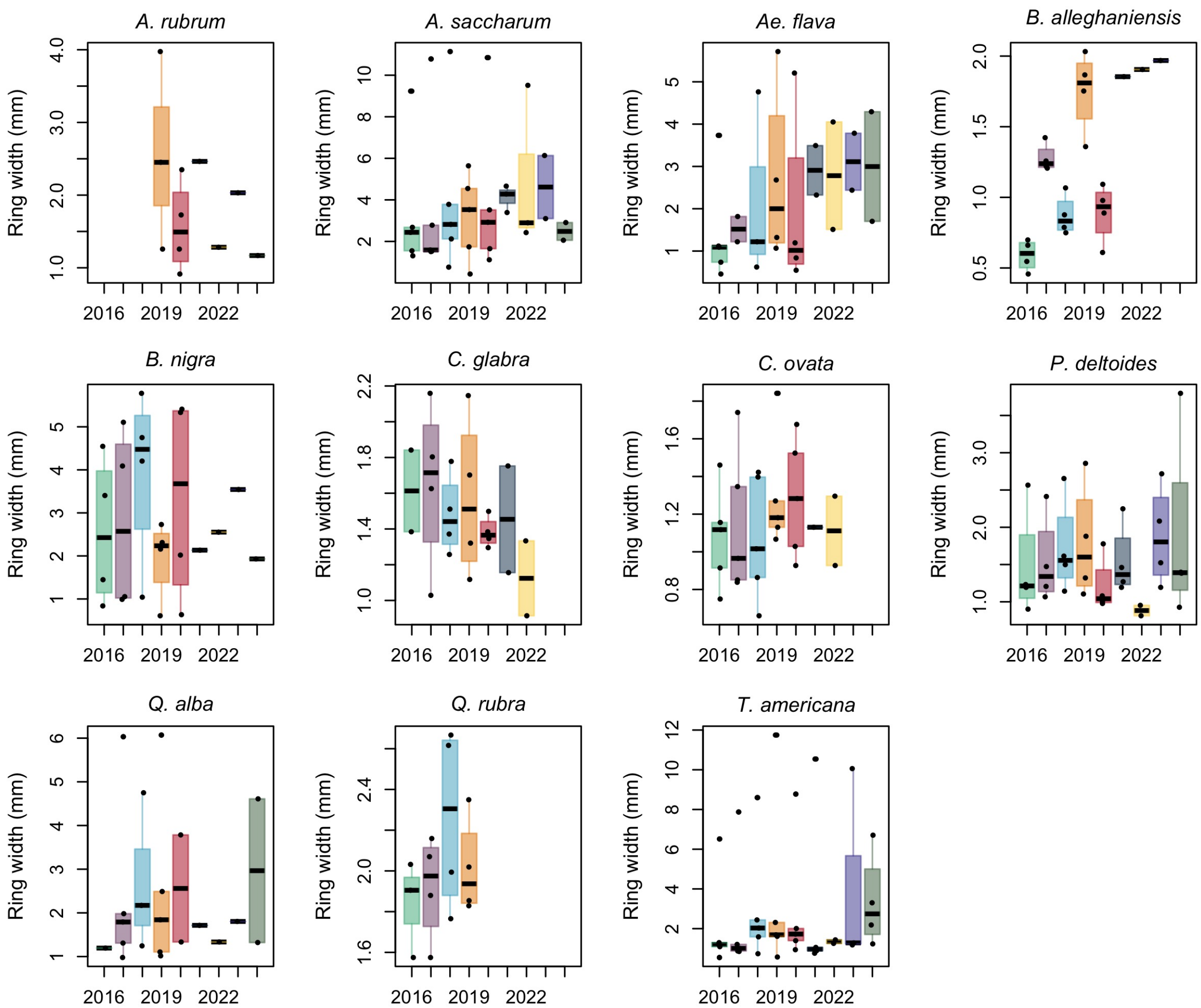

Figure S3: Boxplot representing the empirical data of our ring widths (mm) across years, facetted by species. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

Table S3: Summary output of each parameter from our four predictors, using ring width (log(mm)) as the response variable (See equation 1). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' section in Methods for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{species_{A.rubrum}}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{species_{A.saccharum}}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{species_{Ae.flava}}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{species_{B.alleghaniensis}}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{species_{B.nigra}}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{species_{C.glabra}}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{species_{C.ovata}}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{species_{P.deltoides}}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{species_{Q.alba}}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{species_{Q.rubra}}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{species_{T.americana}}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year_{2016}}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{year_{2017}}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{year_{2018}}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{year_{2019}}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{year_{2020}}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{year_{2021}}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{year_{2022}}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{year_{2023}}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{year_{2024}}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{species_{A.rubrum}}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{species_{A.saccharum}}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]
$\alpha_{species_{Ae.flava}}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{species_{B.alleghaniensis}}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{species_{B.nigra}}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{species_{C.glabra}}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{species_{C.ovata}}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{species_{P.deltoides}}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{species_{Q.alba}}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{species_{Q.rubra}}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{species_{T.americana}}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]

Table S4: Summary output of each parameter from the previous year model (Equation 2) using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	0.20 [-1.10, 1.48]
$\sigma_{\alpha_{tree}}$	0.58 [0.47, 0.72]
σ_y	0.29 [0.26, 0.32]
$\beta^{\text{current}}_{\text{species}_{A.rubrum}}$	-0.07 [-0.25, 0.12]
$\beta^{\text{current}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{current}}_{\text{species}_{Ae.flava}}$	0.04 [-0.02, 0.09]
$\beta^{\text{current}}_{\text{species}_{B.alleghaniensis}}$	-0.15 [-0.29, -0.01]
$\beta^{\text{current}}_{\text{species}_{B.nigra}}$	0.12 [0.03, 0.21]
$\beta^{\text{current}}_{\text{species}_{C.glabra}}$	-0.01 [-0.18, 0.15]
$\beta^{\text{current}}_{\text{species}_{C.ovata}}$	-0.06 [-0.20, 0.08]
$\beta^{\text{current}}_{\text{species}_{P.deltoides}}$	-0.01 [-0.10, 0.08]
$\beta^{\text{current}}_{\text{species}_{Q.alba}}$	0.01 [-0.32, 0.32]
$\beta^{\text{current}}_{\text{species}_{Q.rubra}}$	0.08 [-0.14, 0.32]
$\beta^{\text{current}}_{\text{species}_{T.americana}}$	-0.05 [-0.12, 0.03]
$\beta^{\text{previous}}_{\text{species}_{A.rubrum}}$	0.05 [-0.15, 0.25]
$\beta^{\text{previous}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{previous}}_{\text{species}_{Ae.flava}}$	-0.01 [-0.07, 0.04]
$\beta^{\text{previous}}_{\text{species}_{B.alleghaniensis}}$	0.15 [0.02, 0.28]
$\beta^{\text{previous}}_{\text{species}_{B.nigra}}$	-0.08 [-0.17, -0.00]
$\beta^{\text{previous}}_{\text{species}_{C.glabra}}$	0.01 [-0.13, 0.15]
$\beta^{\text{previous}}_{\text{species}_{C.ovata}}$	-0.04 [-0.17, 0.09]
$\beta^{\text{previous}}_{\text{species}_{P.deltoides}}$	0.02 [-0.05, 0.10]
$\beta^{\text{previous}}_{\text{species}_{Q.alba}}$	-0.03 [-0.38, 0.33]
$\beta^{\text{previous}}_{\text{species}_{Q.rubra}}$	0.09 [-0.27, 0.46]
$\beta^{\text{previous}}_{\text{species}_{T.americana}}$	0.00 [-0.06, 0.06]
$\alpha_{\text{year}_{2016}}$	0.44 [-3.10, 3.90]
$\alpha_{\text{year}_{2017}}$	-2.20 [-4.79, 0.36]
$\alpha_{\text{year}_{2018}}$	0.04 [-1.53, 1.59]
$\alpha_{\text{year}_{2019}}$	-0.06 [-2.61, 2.44]
$\alpha_{\text{year}_{2020}}$	0.30 [-1.39, 1.99]
$\alpha_{\text{year}_{2021}}$	0.18 [-1.77, 2.13]
$\alpha_{\text{year}_{2022}}$	1.16 [-1.18, 3.47]
$\alpha_{\text{year}_{2023}}$	-0.01 [-1.92, 1.90]
$\alpha_{\text{year}_{2024}}$	0.97 [-5.55, 7.68]
$\alpha_{\text{species}_{A.rubrum}}$	-1.73 [-8.12, 4.60]
$\alpha_{\text{species}_{A.saccharum}}$	0.93 [-0.65, 2.50]
$\alpha_{\text{species}_{Ae.flava}}$	0.07 [-0.48, 0.62]
$\alpha_{\text{species}_{B.alleghaniensis}}$	0.06 [-0.50, 0.61]
$\alpha_{\text{species}_{B.nigra}}$	-0.30 [-1.02, 0.42]
$\alpha_{\text{species}_{C.glabra}}$	-0.04 [-0.58, 0.53]
$\alpha_{\text{species}_{C.ovata}}$	0.00 [-0.55, 0.56]
$\alpha_{\text{species}_{P.deltoides}}$	-0.00 [-0.56, 0.56]
$\alpha_{\text{species}_{Q.alba}}$	0.17 [-0.40, 0.73]
$\alpha_{\text{species}_{Q.rubra}}$	0.01 [-0.56, 0.58]
$\alpha_{\text{species}_{T.americana}}$	-0.05 [-0.60, 0.51]

Table S5: Summary output of each parameter from the model on the effect of SOS (days) on EOS (days; Equation 3). We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	29.51 [23.54, 35.39]
$\sigma_{\alpha_{treeid}}$	1.18 [0.83, 1.56]
σ_y	2.06 [1.90, 2.23]
$\beta_{\text{species}_{A.rubrum}}$	0.63 [-0.14, 1.42]
$\beta_{\text{species}_{A.saccharum}}$	0.18 [-0.29, 0.66]
$\beta_{\text{species}_{Ae.flava}}$	0.38 [-0.32, 1.08]
$\beta_{\text{species}_{B.alleghaniensis}}$	0.50 [-0.15, 1.16]
$\beta_{\text{species}_{B.nigra}}$	0.44 [-0.16, 1.06]
$\beta_{\text{species}_{C.glabra}}$	0.48 [-0.17, 1.15]
$\beta_{\text{species}_{C.ovata}}$	0.40 [-0.22, 1.01]
$\beta_{\text{species}_{P.deltoides}}$	1.04 [0.39, 1.70]
$\beta_{\text{species}_{Q.alba}}$	0.35 [-0.34, 1.04]
$\beta_{\text{species}_{Q.rubra}}$	0.50 [-0.19, 1.22]
$\beta_{\text{species}_{T.americana}}$	0.46 [-0.14, 1.05]
$\alpha_{\text{year}_{2016}}$	-2.26 [-15.00, 10.18]
$\alpha_{\text{year}_{2017}}$	7.21 [-1.83, 16.23]
$\alpha_{\text{year}_{2018}}$	0.36 [-10.75, 11.25]
$\alpha_{\text{year}_{2019}}$	-0.23 [-11.38, 10.91]
$\alpha_{\text{year}_{2020}}$	0.27 [-10.43, 10.84]
$\alpha_{\text{year}_{2021}}$	-0.20 [-11.26, 10.69]
$\alpha_{\text{year}_{2022}}$	1.92 [-9.19, 13.03]
$\alpha_{\text{year}_{2023}}$	-11.53 [-23.17, 0.25]
$\alpha_{\text{year}_{2024}}$	3.61 [-8.48, 15.48]
$\alpha_{\text{species}_{A.rubrum}}$	0.14 [-12.16, 12.19]
$\alpha_{\text{species}_{A.saccharum}}$	-0.27 [-10.42, 9.76]
$\alpha_{\text{species}_{Ae.flava}}$	1.86 [-0.38, 4.14]
$\alpha_{\text{species}_{B.alleghaniensis}}$	1.17 [-1.06, 3.41]
$\alpha_{\text{species}_{B.nigra}}$	0.47 [-1.80, 2.72]
$\alpha_{\text{species}_{C.glabra}}$	-0.42 [-2.70, 1.80]
$\alpha_{\text{species}_{C.ovata}}$	-1.02 [-3.27, 1.27]
$\alpha_{\text{species}_{P.deltoides}}$	-0.23 [-2.54, 2.06]
$\alpha_{\text{species}_{Q.alba}}$	-0.87 [-3.21, 1.45]
$\alpha_{\text{species}_{Q.rubra}}$	-0.28 [-2.63, 2.06]
$\alpha_{\text{species}_{T.americana}}$	-1.50 [-3.85, 0.81]

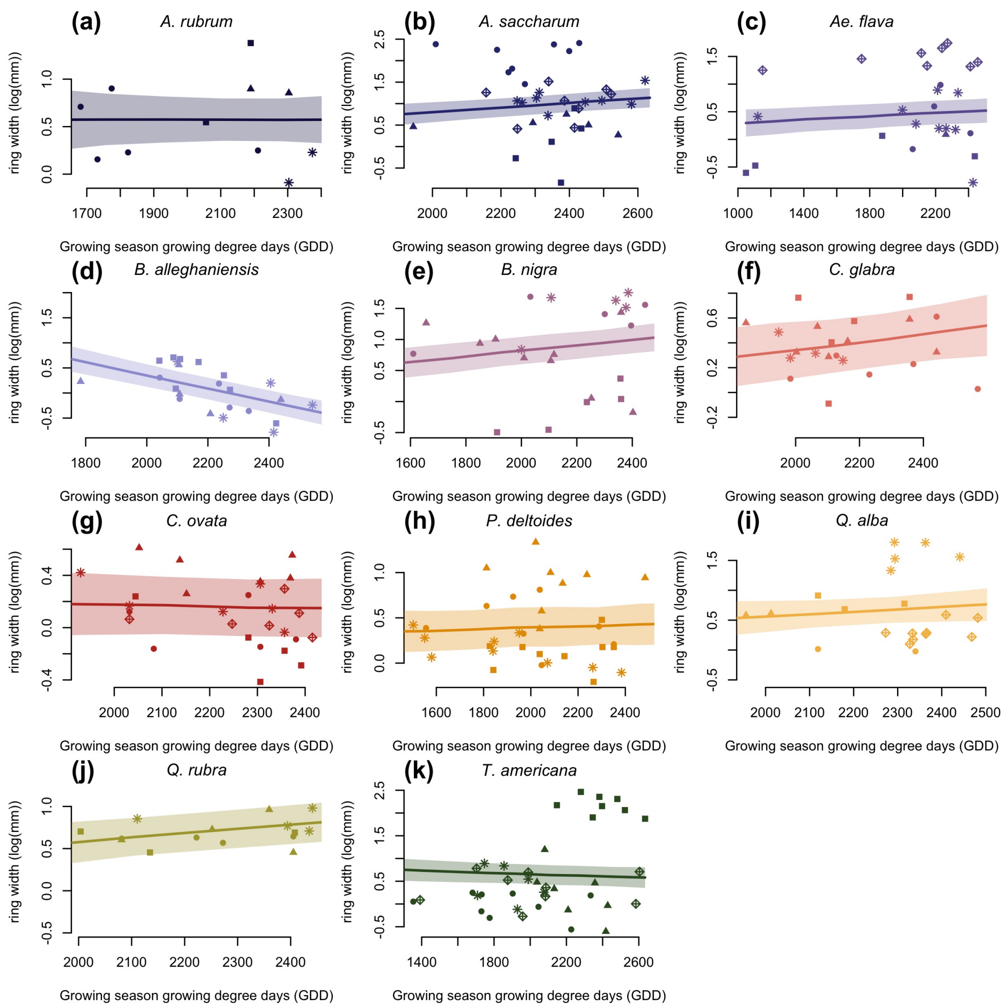

Figure S4: Ring width (log(mm)) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

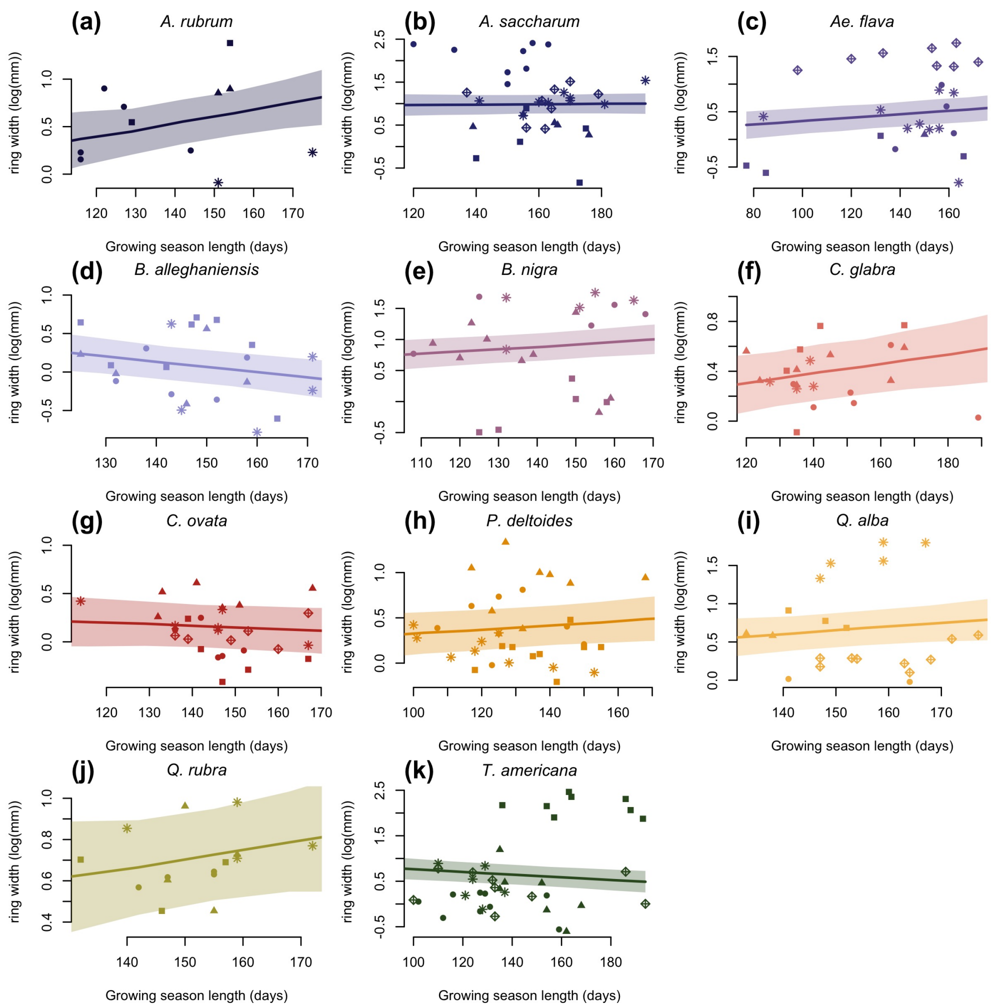

Figure S5: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

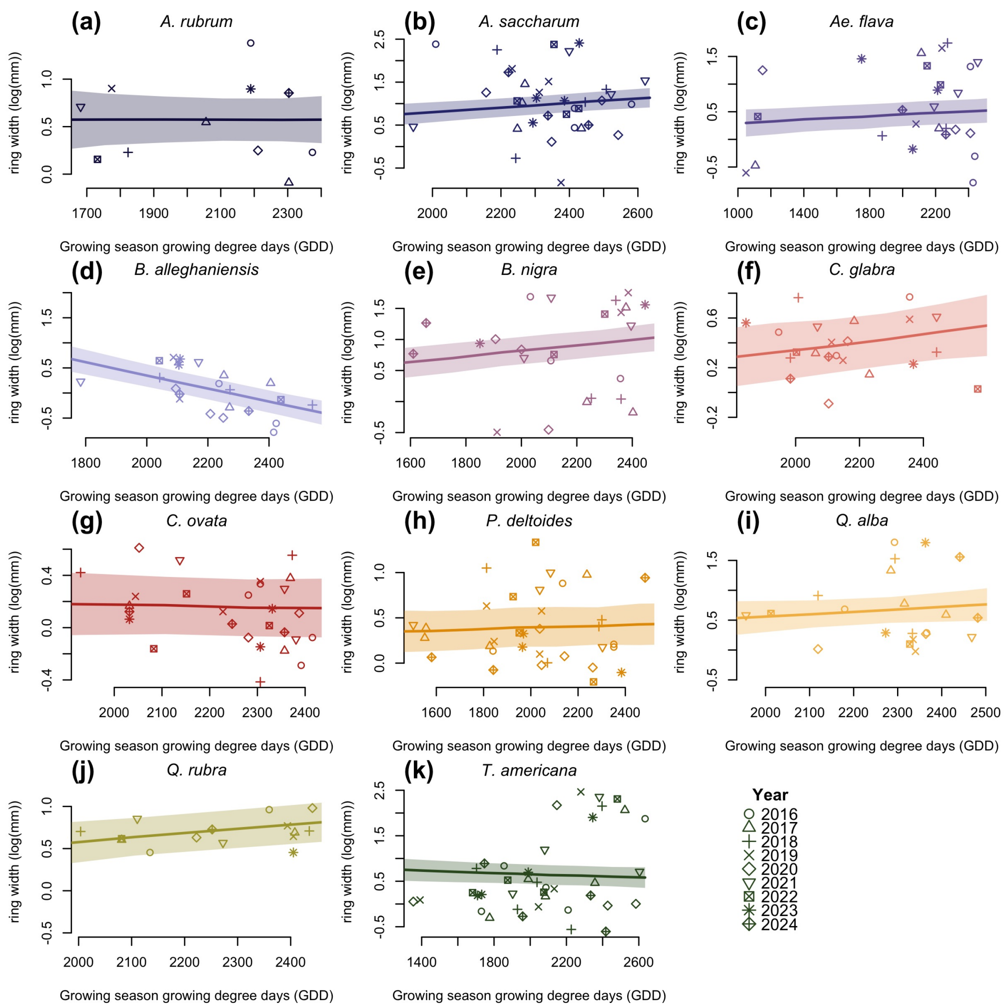

Figure S6: Ring width trends (log(mm)) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

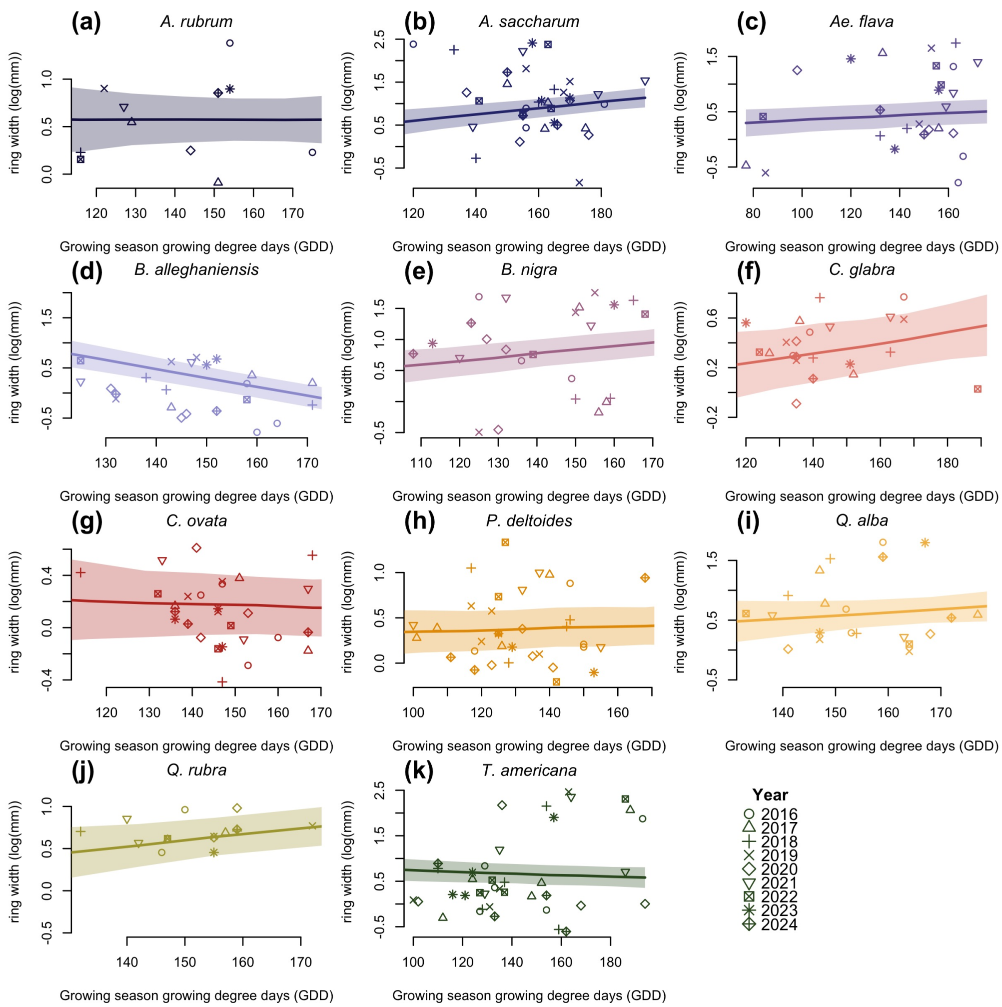

Figure S7: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

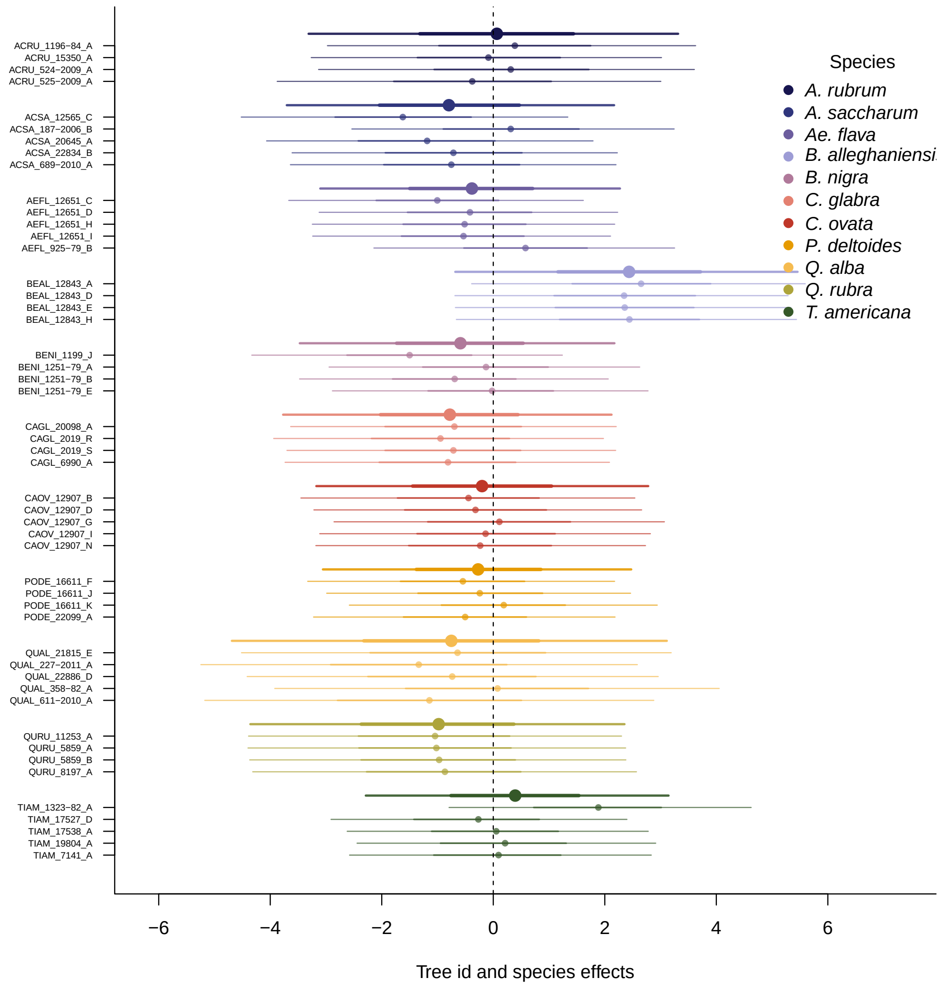

Figure S8: Ring width (log(mm)) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

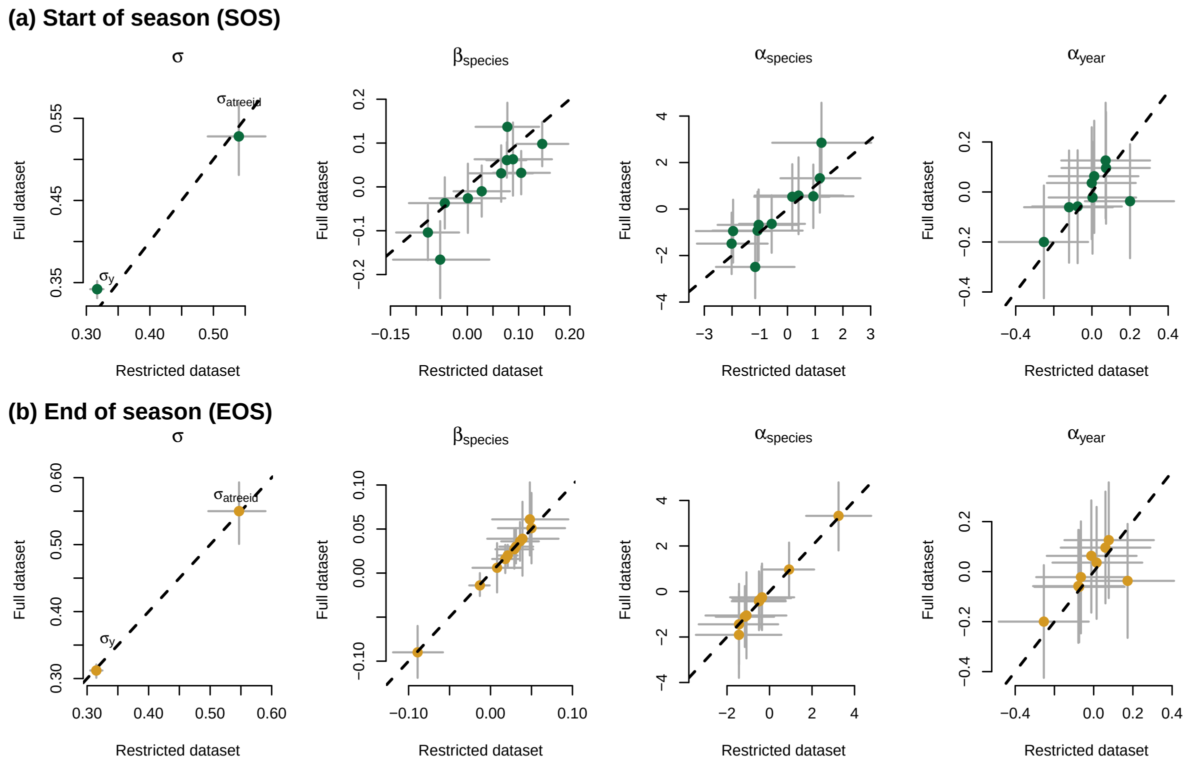

Figure S9: Full vs, most restricted rows of data for the model with start of season (SOS; a) and end of season (EOS; b) as the predictor. Each panel represent the different parameters used to fit the models. $\alpha_{species}$ and α_{year} represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ represent the change in ring width (log(mm)) per scaled predictor for each species (see 'Data analyses' section in Methods for more details on predictor scaling).

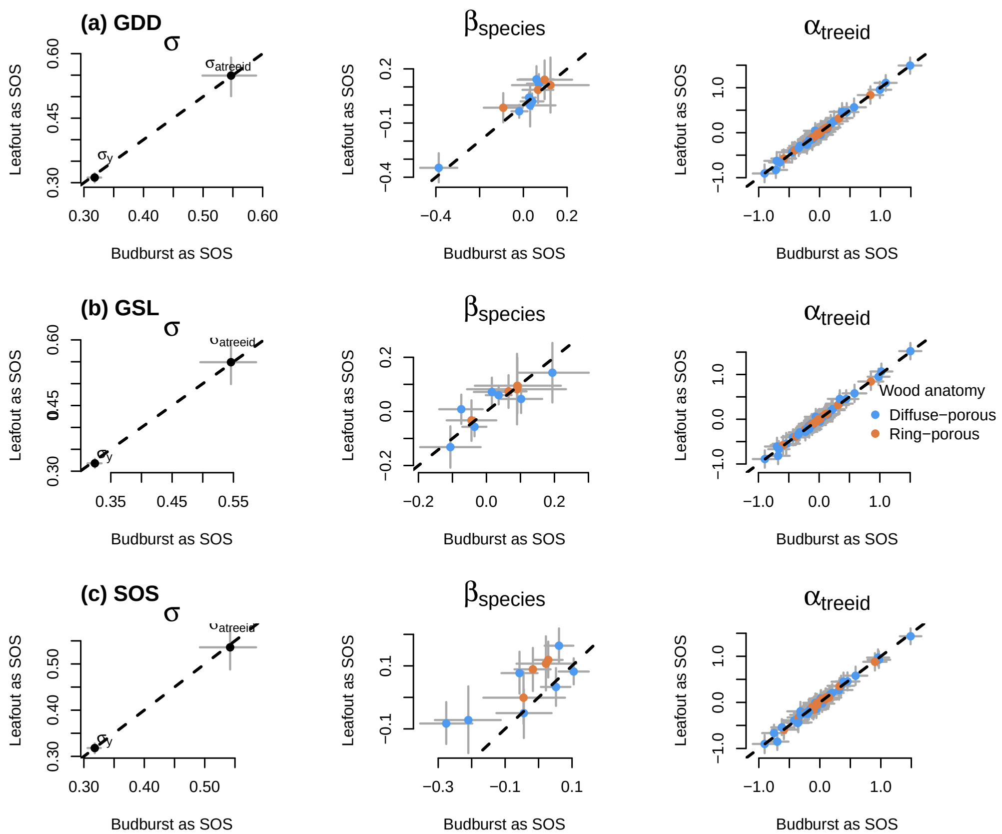

Figure S10: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see ‘Data analyses’ section in Methods for more details).

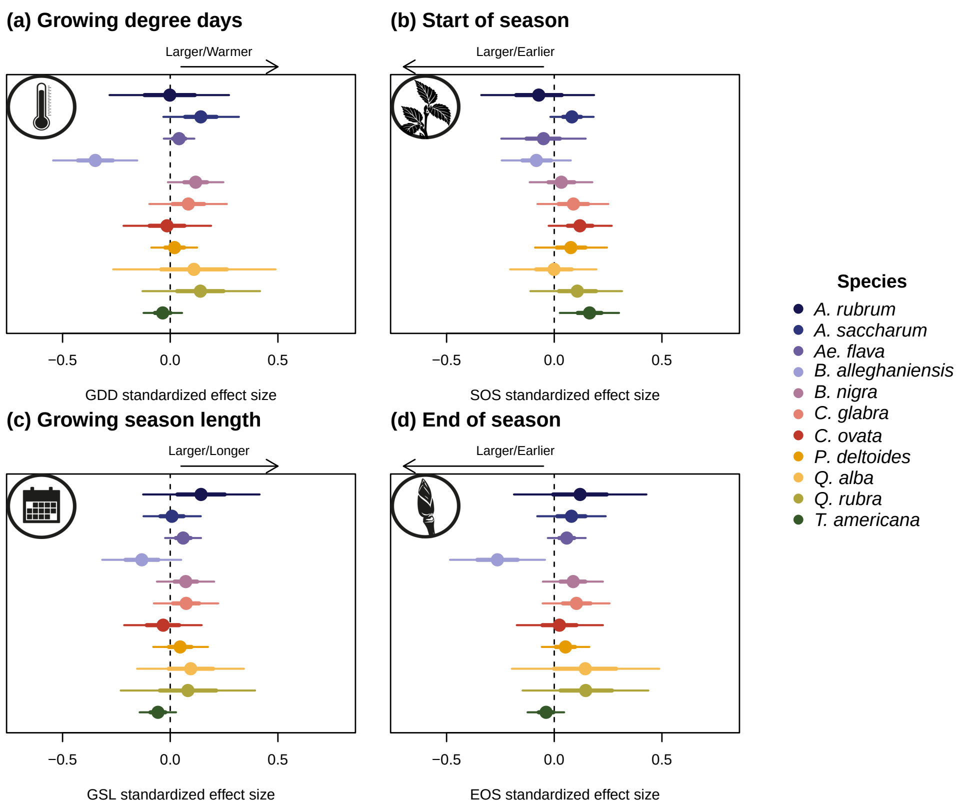

Figure S11: Effect size of each predictors across our eleven species. We depict ring width (log(mm)) change per scaled (z-scored; see 'Data analyses' section in Methods for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

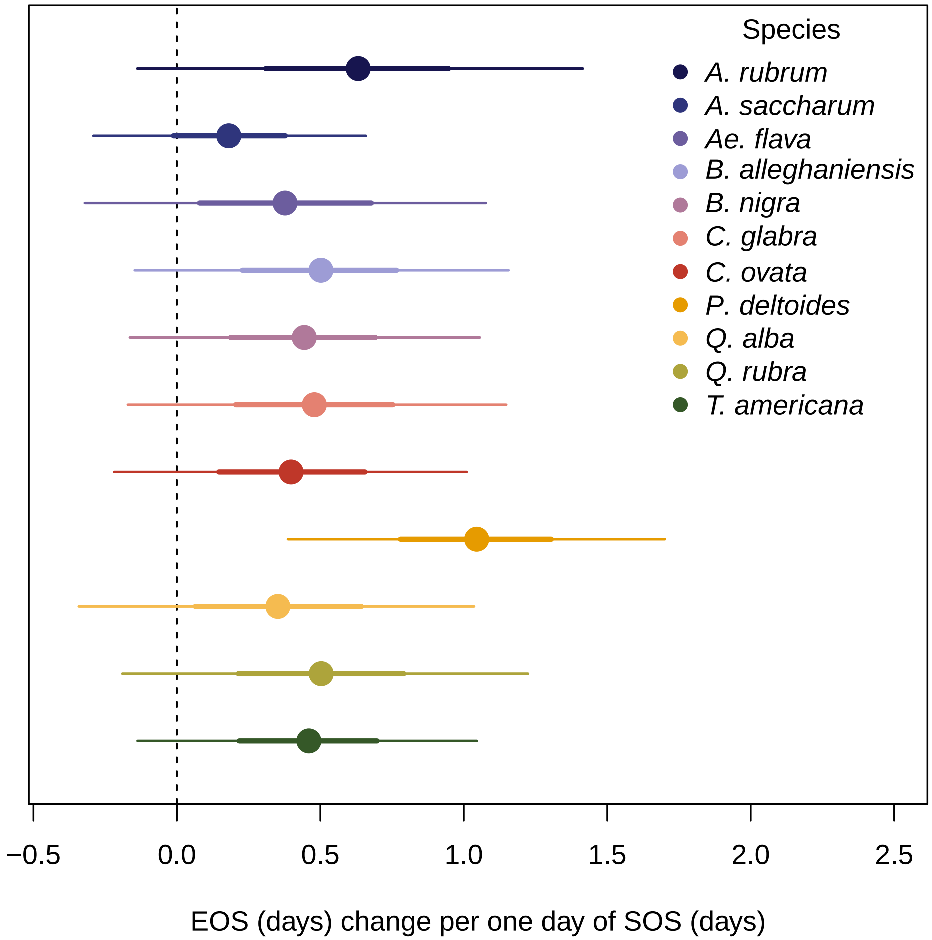

Figure S12: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.